Frequency response of the measuring chain

Ex-situ and in-situ response in the local EIS pipeline: how they are generated, what they show on order 2612030, and what they mean for evaluating and reading the results

Data: R2-D2 gen1 plate (Kashyyyk Abgleich, 72 bode sweeps); runs `snr_0.0_301392ab3c` at 60, 150 and 450 A (67 measured segments, full band 0.19 Hz – 11.95 kHz, fs = 25 kHz). All numbers in this document were recomputed from these files with the pipeline's own module `checks/frequency_response.py`; they match the runs' `plausibility.csv`.

Key messages

- **Ex-situ** = the plate's current amplifiers measured on the bench (Gamry bode sweep per segment). They are transparent below 1 kHz, roll off to −9.9° at 4 kHz, and all 72 are alike within 0.5°: an equivalent delay of 6.9 µs.
- **In-situ** = every segment's impedance divided by the plate median during the real measurement. It sees the whole chain (interface boxes, cables, DAQ cards), but only what differs between channels.
- The large errors are **not** in the amplifiers but downstream: per-channel delays of −58 … +76 µs that turn the phase by up to ±160° at 12 kHz. They are identical at 60, 150 and 450 A (r = 1.000), so they are hardware.
- The timing corrections bring the phase spread between segments at 11.95 kHz from **77° to 1.7°**.
- But a high-frequency bias tied to each channel remains: within a card, R_ohmic drops by **3.3 mΩ·cm² per 10 µs** of removed lag (r = −0.81). Low-frequency results (R_ct, R_mt, R_pol, j_dc) do not show it.
- Practical rule: trust everything below ~500 Hz; treat R_ohmic and the top of the band as provisional until an absolute bench calibration through the full chain exists.

1 Why the measuring chain decides the high-frequency result

The pipeline never measures an impedance directly. For every segment s and every applied tone it forms the ratio of two recorded phasors:

$$Z_s(f) = K_s \cdot U_{UC2}(f) / u_s(f)$$

U_{UC2} is the cell voltage (recorded on every card), u_s the segment's current-channel voltage and K_s its DC calibration from `curr.csv`. Anything the chain does to u_s or U_{UC2} ends up in Z. A chain with response H(f) gives

$$Z_{meas} = Z_{true} / H(f) \text{ and for a phase error } \theta\text{: } \mathrm{Re}\, Z_{meas} \approx \mathrm{Re}\, Z_{true} + \theta \cdot \mathrm{Im}\, Z_{true}$$

So a phase error rotates every point about the origin, and moves the real part by θ·Im Z. θ grows with frequency (a delay τ gives θ = 2πfτ), which is why the error lands on the top of the band – exactly where R_ohmic is read – while the low-frequency arc looks perfect.

Worked example – segment 19, 150 A

Segment 19's current channel lags the plate by τ = 75.8 µs. Using its corrected spectrum and putting the lag back in:

f	phase error θ = 2πfτ	corrected Z [mΩ·cm²]	with the lag [mΩ·cm²]	effect on Re Z
597 Hz	16°	56.4 − 19.0j	48.8 − 34.1j	−7.6 (−13 %)
1.19 kHz	33°	48.2 − 9.4j	35.5 − 33.9j	−12.7 (−26 %)
2.39 kHz	65°	36.4 − 3.9j	11.7 − 34.7j	−24.7
3.80 kHz	104°	27.1 − 0.3j	−6.7 − 26.3j	negative: no intercept

Without the correction the high-frequency intercept of this segment cannot even be read. This is why the pipeline looks at the chain's frequency response twice – once on the bench, once in the measurement – instead of assuming it.

2 The signal chain and what each view can see

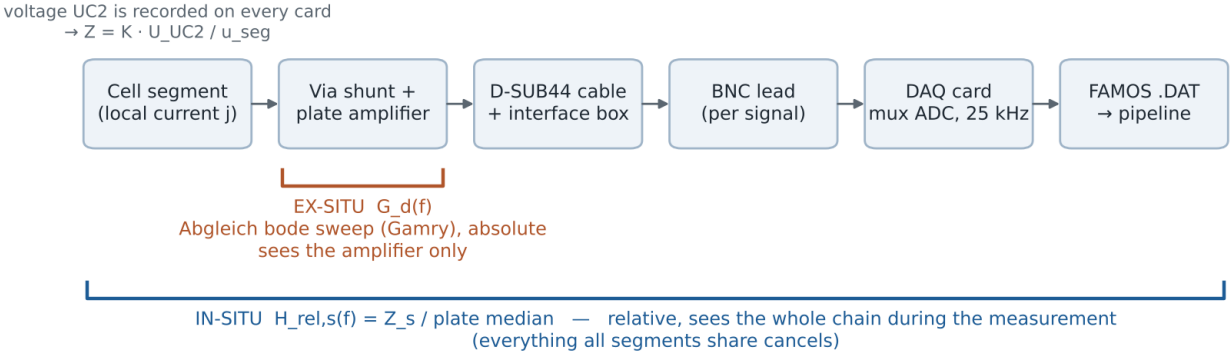


Figure 1 – The current path of one segment. The ex-situ sweep covers only the plate's shunt and amplifier; the in-situ comparison covers everything, including the cell, but only relative to the other segments.

	Ex-situ response $G_d(f)$	In-situ response $H_{rel,s}(f)$
What is measured	each segment's current amplifier, swept by a Gamry (no cell)	each segment's impedance during the real EIS sweep
Definition	$G_d = j_\sigma / j_r$, normalised to 1 in the flat band 5–500 Hz	$H_{rel,s} = Z_s(f)$ / median over segments Z(f)

	Ex-situ response $G_d(f)$	In-situ response $H_{rel,s}(f)$
Absolute or relative	absolute	relative – whatever all segments share cancels
Sees	shunt + plate amplifier	amplifier, D-SUB44, interface box, BNC, DAQ card (multiplexer, inputs), card timing
Blind to	everything after the amplifier output	any error common to all channels; real cell differences also appear in it
When	once per plate (Abgleich campaign)	every run, every operating point
Used for	the chain gain file, divided out in bronze	diagnosis, and to justify the in-situ timing / HF corrections

3 Ex-situ response: how it is generated

Source data

Every Abgleich delivery contains a `bode/` folder with one Gamry file per segment, `<plate>_100kHz_1Hz_500mA_#<n>.DTA`. Each is a galvanostatic sweep of that segment's current-measurement path at 500 mA rms, from 100 kHz down to 1 Hz, with no DC bias and no cell. It is not electrochemistry: it is the shunt plus its amplifier.

Processing (readers/gamry_dta.py, runner cell “Chain response”)

- **Read** the ZCURVE table of every sweep → (f, Z) per segment (`read_bode_folder`).
- **Normalise**: divide by the median of Z over the flat band 5–500 Hz (`z_flat`), so H = 1 where the amplifier is flat. The DC level is already in K from `curr.csv`; only the frequency dependence is wanted.
- **Cross-check the index**: the low-frequency |Z| of each sweep is the same DC path as c0 in `curr.csv`. Gen1 (Kashyyyk): r = +0.999, ratio spread 0.6 % → sweep #n is segment n. Gen2 (Naboo): r = +0.41 → the numbering does not match, so the runner writes the plate median instead (‘shared’ mode).
- **Export** the gain file `EIS_config/<plate>/chain_gain.csv` with gain = 1/H (the factor Z is divided by). Bronze applies it to every segment before anything else: Z ← Z / gain.
- **Report** (checks/frequency_response.py): G_d per segment, the values at 100 Hz, 1, 4 and 10 kHz, the segment spread, and the plausibility row “chain response (ex-situ)”.

Plausibility rules: every sweep flat within 1 % between 5 and 500 Hz (else a step at the normalisation point would be carried into Z at every frequency); segment spread of the phase at 4 kHz ≤ 2°.

Result on the gen1 plate

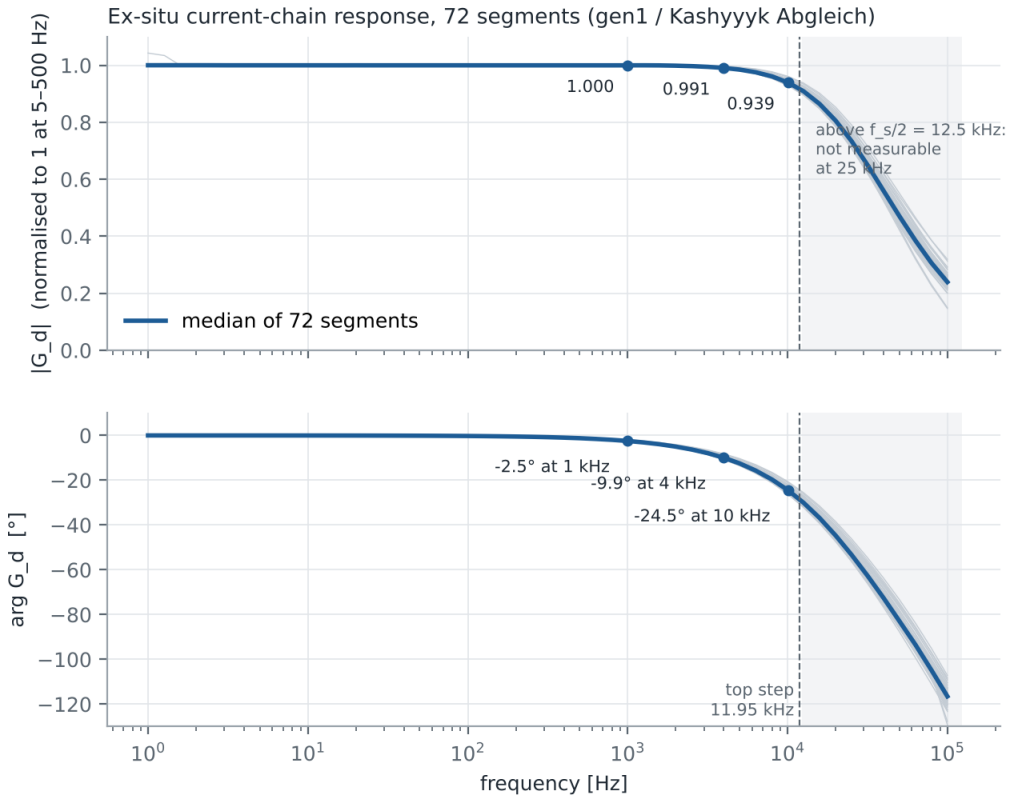


Figure 2 – Ex-situ response of the 72 current amplifiers (grey) and their median (blue). Shaded: above half the sampling rate, never measured by the FAMOS cards.

f	$\|G_d\|$ median	arg G_d median	segment spread (sd)
100 Hz	1.000	−0.2°	0.02°
1 kHz	1.000	−2.5°	0.15°
4 kHz	0.991	−9.9°	0.52°
5 kHz	0.985	−12.5°	0.64°
10 kHz	0.939	−24.5°	1.13°
31.6 kHz	0.651	−62.6°	2.4°
100 kHz	0.240	−116.5°	3.9°

Read as a delay, the phase slope between 50 Hz and 4 kHz is 6.9 µs for every segment (range 5.7–7.5 µs). All sweeps are flat within 1 % in 5–500 Hz. Verdict at 60, 150 and 450 A: **PASS**.

4 In-situ response: how it is generated

The in-situ view needs no extra measurement. It uses the fact that all segments sit on one cell, so their true spectra have a similar shape. Dividing each segment by the plate median removes the cell and leaves what is different about that segment's channel (plus any real local difference).

$$H_{rel,s}(f) = Z_s(f) \,/\, \text{median}_{segments}\, Z(f) \text{ (median of real and imaginary parts separately)}$$

Processing (checks/frequency_response.py, runner cell “Frequency response”)

- **Two stages on the same points**: raw from `bronze/raw_spectra.csv` (chain gain file divided out, no timing correction) and clean from `silver/spectra_clean.csv` (after all corrections). Only the points silver kept are used, so the two stages are directly comparable.
- A frequency is used only if at least half of the segments have a value there – a median of three is not a plate reference.

- **Phase spread**: standard deviation over segments of arg H_{rel} at 100 Hz, 1 kHz, 2 kHz and the top step.
- **Card fingerprint**: per card, the median phase above 2 kHz and how far $|H_{rel}|$ above 2 kHz moves from the same segment's own 30–1000 Hz level. A flat scale factor does not count; a frequency-dependent card response does.
- **Plausibility** “segment frequency response (in-situ)”: phase sd at the top > 10° → FAIL, > 5° → WARN; any card step > 5 % → WARN.
- Outputs in `RUN_DIR/frequency_response/`: insitu_response.csv, insitu_card_residual.csv, summary.json and the two PNGs. Nothing here changes Z – it is a diagnostic.

Result on 2612030

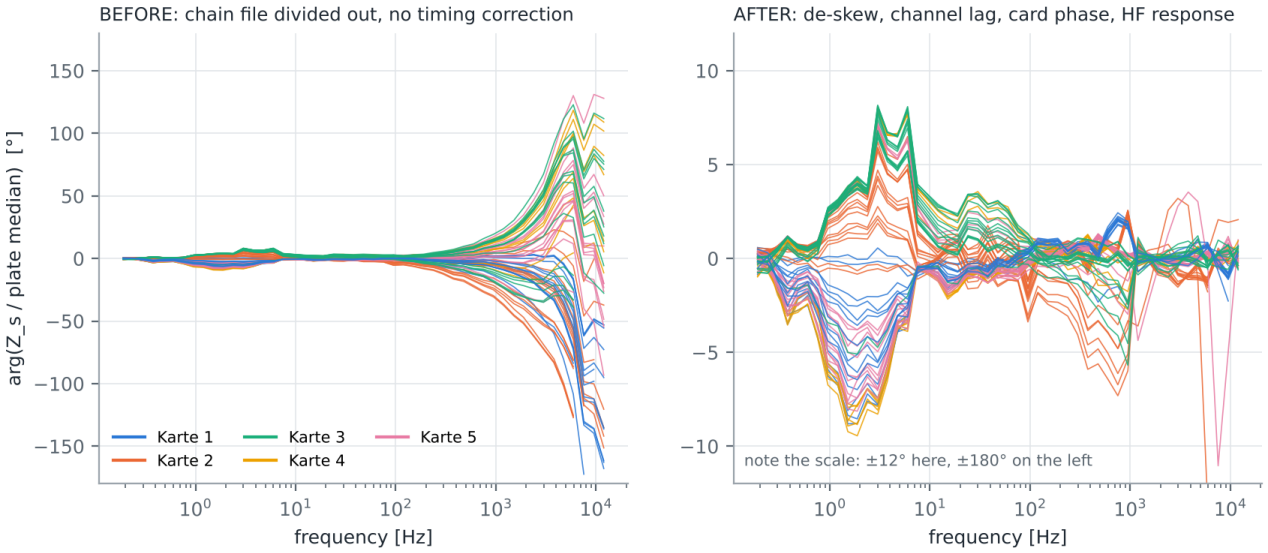


Figure 3 – Phase of every segment against the plate median, 150 A, coloured by DAQ card. Left: before the timing corrections. Right: after them (note the 15× smaller scale).

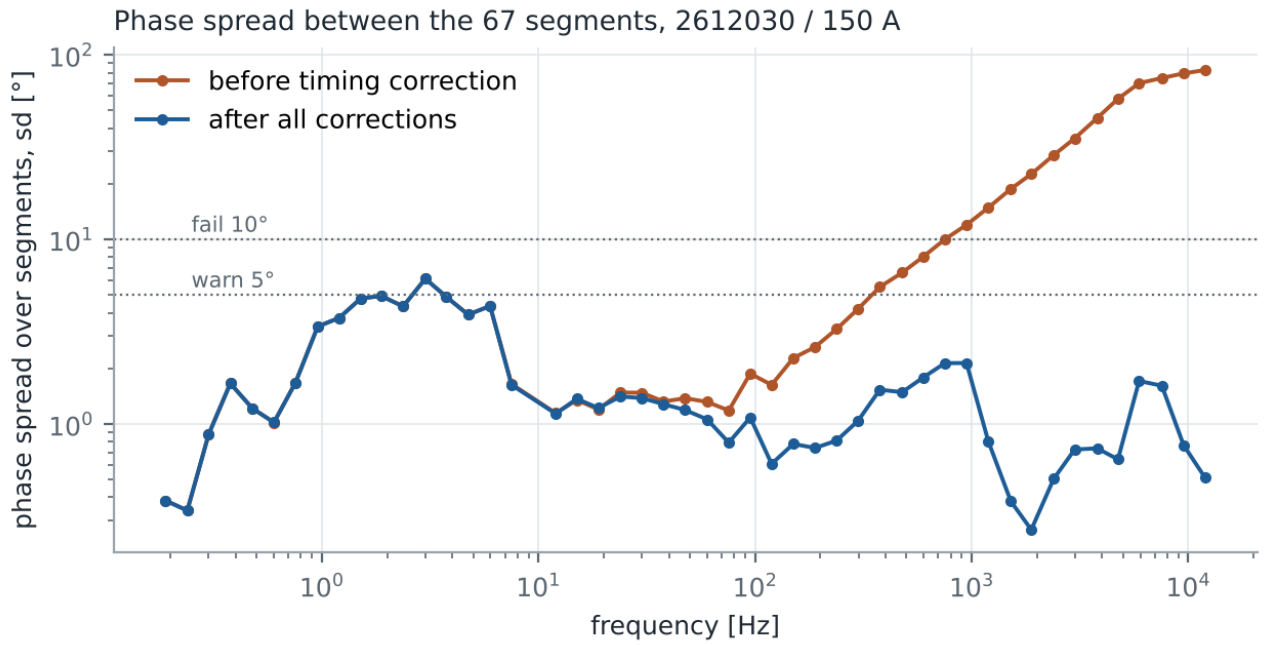


Figure 4 – The same as one number per frequency: the phase spread between segments. Before the corrections it grows linearly with frequency (a delay); after them it stays below 2.2° above 10 Hz.

	60 A	150 A	450 A
phase sd at 11.95 kHz, before	77.9°	76.8°	77.1°
phase sd at 11.95 kHz, after	1.3°	1.7°	2.1°
card |H| step above 2 kHz (K1 … K5)	+2.2 / +2.3 / −3.1 / +3.0 / +2.6 %	+1.4 / +4.7 / −1.8 / +1.5 / +2.2 %	−0.4 / **+6.5** / −1.3 / +0.5 / −0.1 %
verdict	PASS	PASS	WARN (card 2)

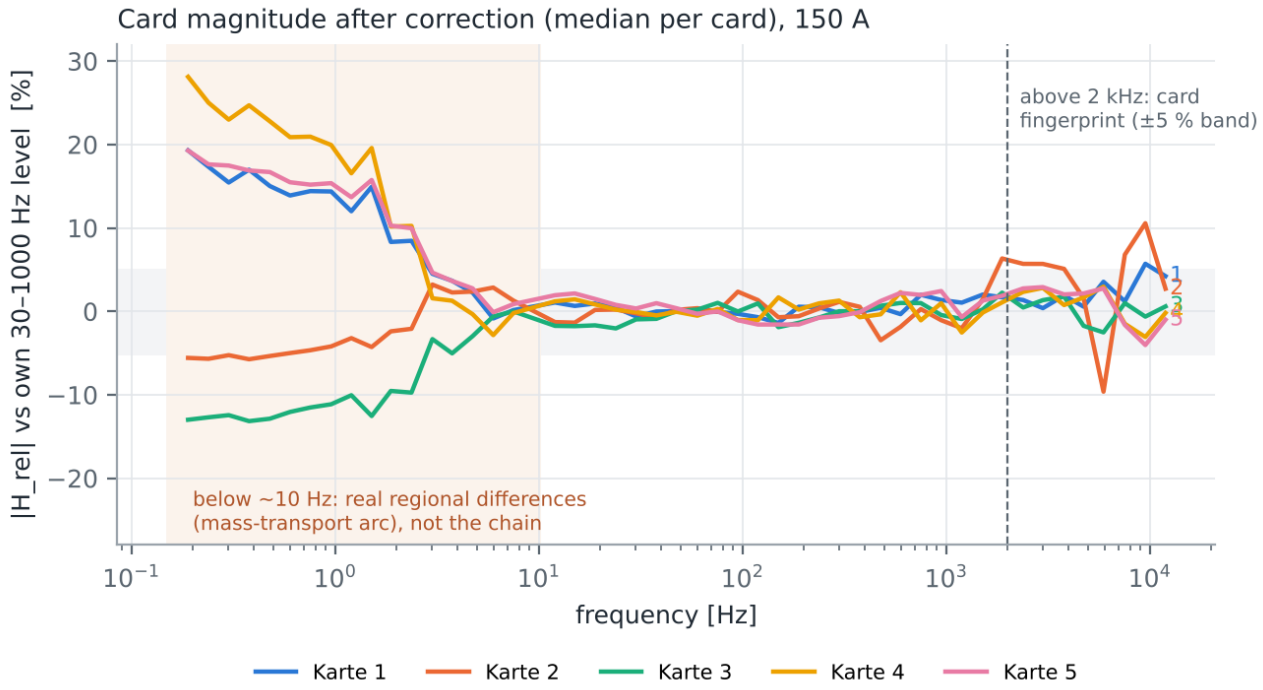


Figure 5 – Magnitude after correction, median per card, relative to each segment's own 30–1000 Hz level. Above 2 kHz each card keeps a small fingerprint (mostly within ±5 %). Below ~10 Hz the cards separate by up to ±25 %: that is the cell – the cards cover different regions of the plate and the mass-transport arc differs between inlet and outlet – and must not be corrected away.

5 Comparing the two views: where the error really sits

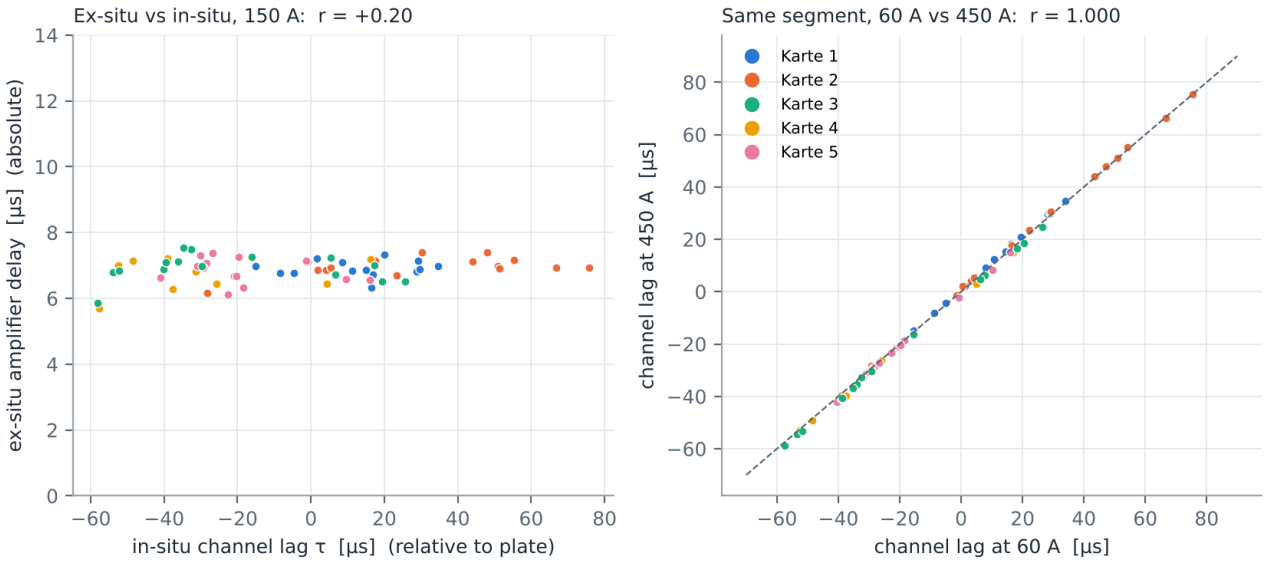


Figure 6 – Left: the ex-situ amplifier delay (6–7.5 µs, the same for every segment) does not explain the in-situ channel lag (−58 … +76 µs). Right: the in-situ lag of every segment is the same at 60 and 450 A.

Three conclusions follow, and they are the core of this document:

- **The amplifiers are not the problem.** Ex-situ, all 72 current amplifiers are alike within 0.4 µs. The in-situ lags spread over 134 µs and correlate only weakly with the ex-situ delay (r = +0.20). The lag enters after the plate: interface boxes, cabling and the DAQ side.
- **The lag is hardware, not electrochemistry.** The cell changes a lot between 60 and 450 A; the lag of each segment does not (r = 1.000, median difference 0.5–0.9 µs). Part of it is per card (card medians +16, +30, −33, −38, −22 µs), part per channel (sd 15–28 µs inside a card); it does not follow the multiplexer slot order.
- **Each view needs the other.** Ex-situ is absolute but blind to the cards; in-situ sees the cards but only relative to the plate. The absolute anchor for what all channels share at high frequency comes from a third source: the whole-cell Gamry sweep (per-card phase above 1 kHz, about 10–13 µs).

How the corrections use the two views (order of application)

#	Correction	Based on	Removes
1	Chain gain file (bronze)	ex-situ bode sweeps	amplifier roll-off: −2.5° at 1 kHz, −9.9° at 4 kHz
2	Multiplexer de-skew (silver)	assumed ADC slot order, 2.5 µs/slot	slot-dependent sampling time
3	Card UC2 gain and HF magnitude	UC2 recorded on all cards	voltage-chain gain and pickup per card
4	Channel lag (silver, channel_lag.py)	in-situ, against the plate median	per-segment delay, −58 … +76 µs
5	Per-card phase (silver)	whole-cell Gamry sweep	phase all channels of a card share, above 1 kHz
6	Frozen HF response (hf_response.py)	in-situ, across all conditions, validated leave-one-out	remaining per-segment phase line and per-card magnitude above 1 kHz

The HF response (6) is built from several operating points, checked on a current it was not fitted on, and then frozen. It is rebuilt only when HF_RESPONSE_REBUILD = True – rebuilding it on every run changed the cache key each time, which was the cause of the ‘cache mode always recomputes’ problem.

6 What remains: a high-frequency bias tied to each channel

The corrections align the phase. They do not prove that the corrected spectrum is right in magnitude at high frequency. A direct test: if the corrections were complete, a segment's R_ohmic would not depend on how much lag was removed from its channel. To separate this from geography (each card covers one region of the plate), the comparison is made inside each card, with position trends removed.

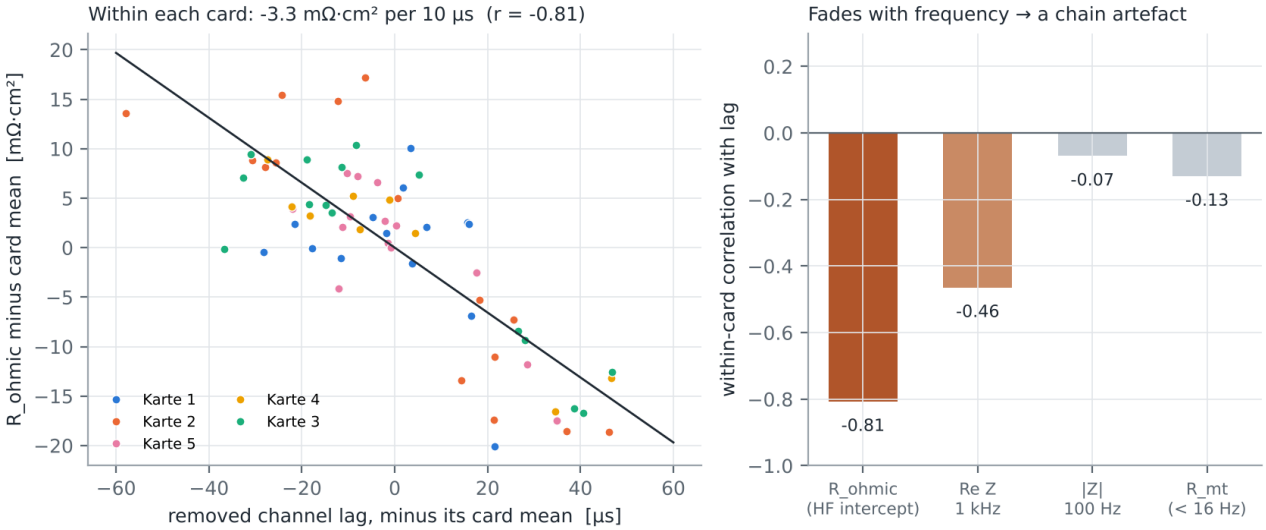


Figure 7 – Left: within each card, segments whose channel needed more lag correction read a lower R_ohmic. Right: the same correlation for quantities read at lower frequencies – it fades with frequency.

Quantity (150 A)	read at	within-card r with lag	slope
R_ohmic	HF intercept (> 2 kHz)	−0.81	−3.3 mΩ·cm² per 10 µs (−6.3 %)
Re Z at 1 kHz	1 kHz	−0.46	−1.4 % per 10 µs
|Z| at 100 Hz	100 Hz	−0.07	−0.2 % per 10 µs
R_mt	< 16 Hz (DRT)	−0.13	–

The slope is the same at 60, 150 and 450 A (−3.2, −3.3, −3.1 mΩ·cm² per 10 µs), and survives removing position trends inside each card (r = −0.47 … −0.90 per card). A correlation that is strongest at the highest frequency and absent at low frequency is the signature of the measuring chain, not of the membrane. With lags spanning 134 µs, the possible bias reaches about ±20 mΩ·cm² – larger than the real R_ohmic spread across the plate (sd 10.7 mΩ·cm²). Segment 19 is the extreme: R_ohmic 26 mΩ·cm² against 53 for its neighbour 14 and a plate median of 52.

Most likely cause: the channels are not pure delays. A delay model aligns the phase but a channel that also changes magnitude with frequency (an input filter, a cable resonance, pickup) leaves its magnitude error in place, and the R_ohmic estimate reads it. Only an absolute calibration through the full chain can separate the two.

7 What we learn

For the EIS evaluation

- **Below ~500 Hz the chain is transparent.** $|G_d| = 1.000$, phase below 1°, amplifiers identical to 0.02°, in-situ spread ≈ 1°. Everything built on the low-frequency part – the DRT, R_ct, R_mt, R_pol, the arcs, j_dc – needs no chain correction and is the most reliable output of the pipeline.
- **The amplifiers need only a common correction.** All 72 are alike within 0.5° at 4 kHz, so one median response would do almost as well as 72 individual ones. That is why gen2, whose bode numbering does not match its calibration, can safely use the plate median.
- **The real high-frequency problem is downstream of the plate.** Per-channel delays of up to ±76 µs, the same at every current, must be removed on every run. Without them the band above ~300 Hz is unusable (phase spread 12° at 1 kHz, 77° at 12 kHz).
- **A relative correction needs an absolute anchor.** In-situ can only bring channels onto the plate median; the Gamry sweep supplies what all channels share. Keep the Gamry sweep in every measurement.
- **The usable top of the band is set by the chain, not by the sampling rate.** 12.5 kHz is the hard limit (Nyquist); in practice, phase-aligned data are good to about 2 kHz, and between 2 and 12 kHz each card keeps a ±5–10 % magnitude fingerprint.
- **It explains why lin-KK fails above 1 kHz.** The KK residual of single segments is dominated by the top of the band, where the chain residue sits; the Gamry spectrum of the same cell passes there.

For analysing and presenting the results

- **Treat the R_ohmic map as provisional.** Compare it with the `chain_tau_us` map: where R_ohmic follows the lag, it shows wiring, not membrane. Report the within-card correlation alongside the map.
- **Rules that rely on R_ohmic inherit this.** The ‘drying’ and ‘contact loss’ labels use ρ(R_ohmic); a low-lag or high-lag channel can push a segment across those thresholds.
- **Low-frequency maps can be read physically.** R_ct, R_mt, R_pol, j_dc and temperature show no link to the channel lag.
- **Patterns shaped like a card or an interface box are suspicious at high frequency.** Each card covers a block of the plate; a step at a card border in an HF map is more likely the card than the cell.
- **Do not ‘correct’ the low-frequency part of the in-situ ratio.** Below ~10 Hz the cards differ by ±25 % because the cell differs (inlet vs outlet). This is why the HF correction is per card in magnitude and starts only at 1 kHz.
- **Read the two plausibility rows on every run.** “chain response (ex-situ)” confirms the gain file belongs to this plate; “segment frequency response (in-situ)” confirms the timing corrections worked (sd ≤ 5°) and flags a card that drifts (450 A: card 2 +6.5 %).
- **When comparing with the Gamry HFR, expect a gap.** The matched-band R_s closure is −5 … −11 %; part of it is this residual chain effect, part the unmeasured 4 % of the plate.

Recommended next steps

Step	What it gives
Bench sweep through the full chain: one signal into all inputs via interface boxes, BNC and DAQ cards	absolute H(f) per channel, magnitude and phase – replaces the relative in-situ corrections and settles the multiplexer question
Add a plausibility check ‘R_ohmic vs channel lag (within card)’	flags automatically when the R_ohmic map carries the wiring
Fit a general per-channel response (delay + magnitude slope) instead of a pure delay	removes the magnitude part the delay model leaves
Read R_ohmic at the highest frequency that is validated, and give its uncertainty from the chain residue	an honest error bar on the HFR map

Appendix – where to find it

Item	Location
Ex-situ reader, gain-file export, index cross-check	readers/gamry_dta.py (read_bode_folder, cross_check_abgleich, write_gain_csv, median_response)
Gain file applied in bronze	EIS_config/<plate>/chain_gain.csv (gain = 1/H; Z ← Z / gain)
Both views, checks, plots	checks/frequency_response.py → RUN_DIR/frequency_response/
Per-channel lag	pipeline/channel_lag.py → silver/channel_lag.csv, map chain_tau_us
Per-card phase against the Gamry	pipeline/silver.py (gamry_card_phase)
Frozen HF response	checks/hf_response.py → EIS_Results/<order>/hf_response/
Runner cells	“Chain response” (build gain file), “Frequency response of the measuring chain”, “High-frequency response across conditions (frozen correction)”
Plausibility rows	plausibility.csv: chain response (ex-situ); segment frequency response (in-situ); channel lag

Glossary

Term	Meaning
$G_d(f)$	ex-situ response of one segment's current amplifier, normalised to 1 at 5–500 Hz
$H_{rel,s}(f)$	in-situ response: segment impedance divided by the plate median at the same frequency
Gain file	per-segment table of 1/H(f) that bronze divides every impedance by
Channel lag τ	delay of one segment's channel relative to the plate, fitted from arg $H_{rel} = -2\pi f\tau$
Phase spread	standard deviation of arg H_{rel} over all segments at one frequency
Card fingerprint	how a card's $\lvert H_{rel}\rvert$ above 2 kHz differs from its own 30–1000 Hz level
Flat band	5–500 Hz, where every amplifier is flat; used for normalisation
Nyquist limit	half the sampling rate, 12.5 kHz at 25 kHz: nothing above can be measured